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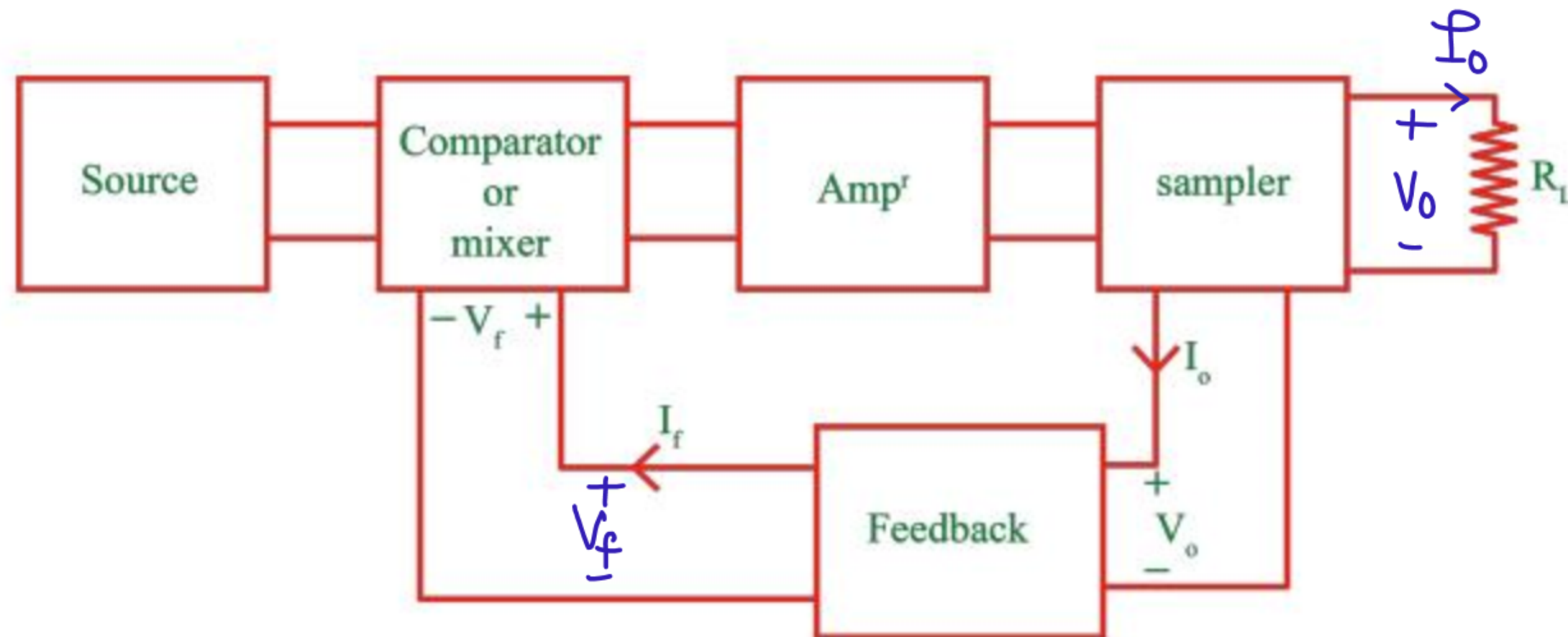
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Feedback Amplifier

Feedback Amplifier

- A feedback amplifier is an amplifier in which the output signal is sampled. And it is fed back to the input to form an error signal which drives the amplifier.



QUESTION

Why negative feedback is used ?

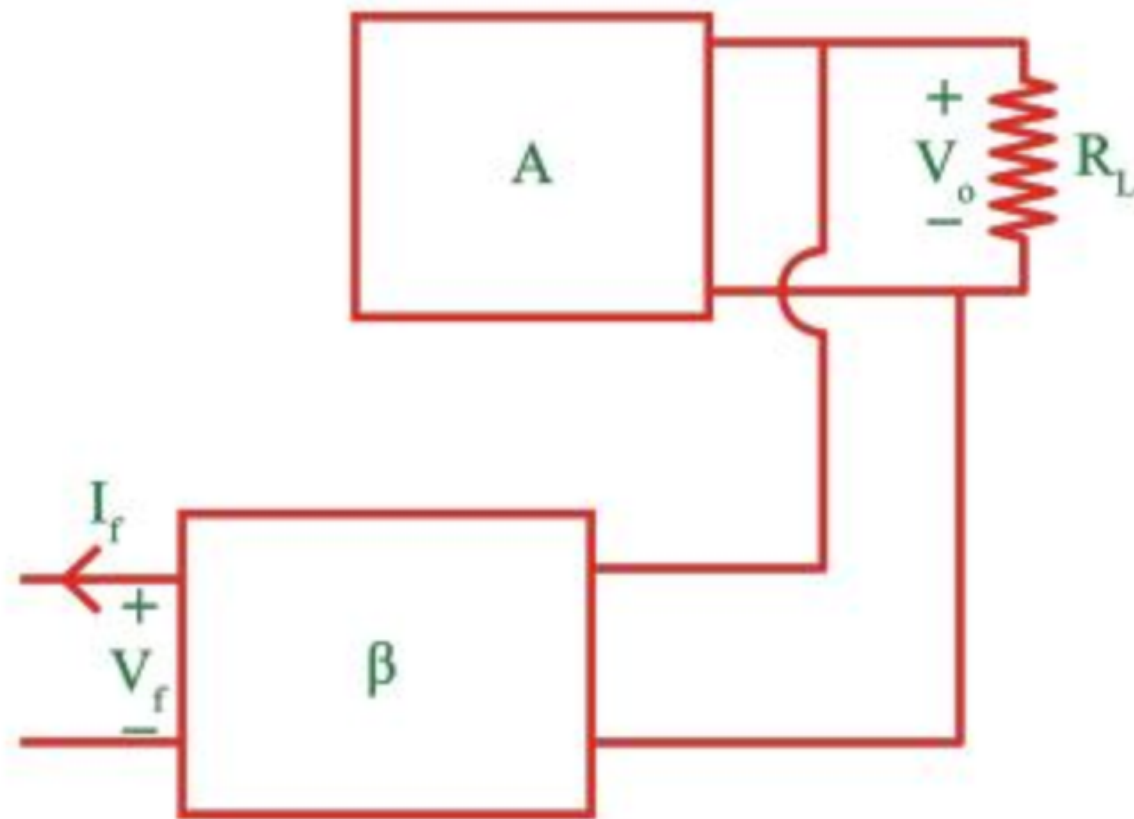
Answer

1. To Stabilize the voltage gain ✓ (Similar to op-amp)
 2. To increase the B.W [gain ↓ BW ↑]
 3. To increase the input impedance
 4. To reduce out put impedance
 5. To reduce noise
- ← for voltage amp^r

Sampling Circuit :

- This circuit is used to sample the output and connect it to feedback network.
- There are two types of sampling circuit :

(1) Shunt Sampling or Voltage Sampling



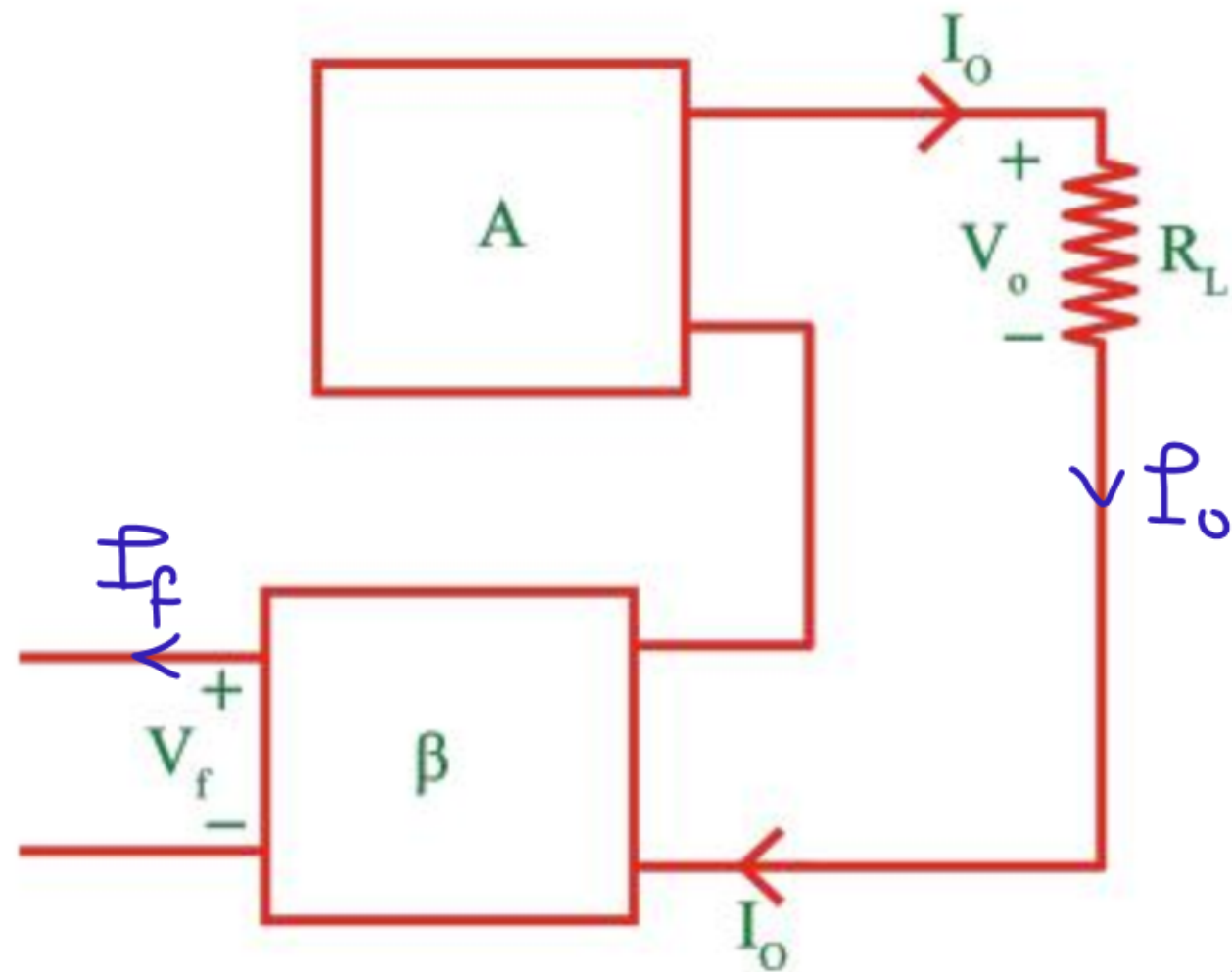
◦ Signal input to feedback n/ω is V_o

◦ feedback factor (β)

$$\beta = I_f / V_o \text{ (}\sigma\text{)}$$

$$\beta = V_f / V_o \text{ (unitless)}$$

(2) Current Sampling or Series ~~of~~ Sampling



◦ Sampler is connected in series with o/p

◦ Sampler picks I_o & feeds to the feedback n/w.

◦ feedback factor (β)

$$\frac{V_f}{I_o} = \beta \text{ (unitless)}$$

$$\beta = \frac{V_f}{I_o} (\Omega)$$

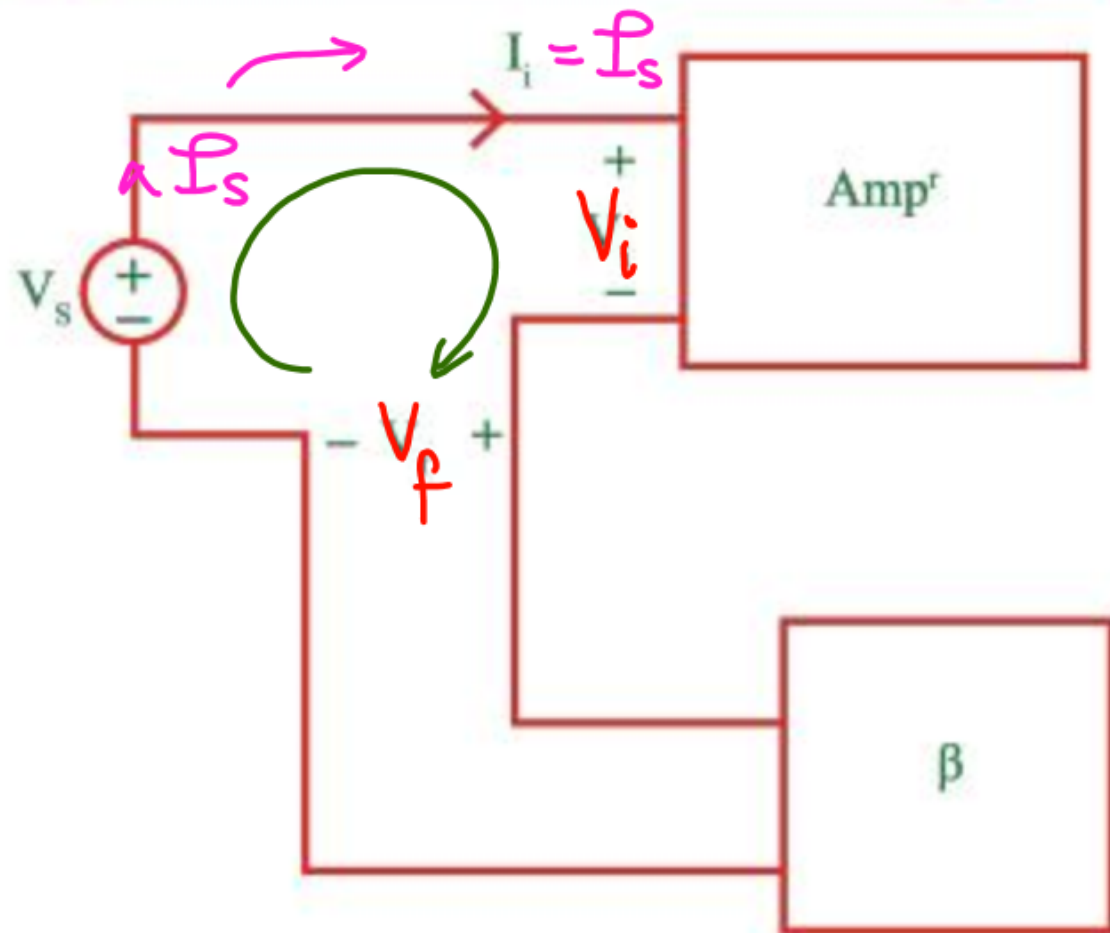
Mixer Network :

- There are two type of mixer network:

(1) Series Mixing :

or voltage mixing

→ The o/p of feedback n/w is connected in series with source.

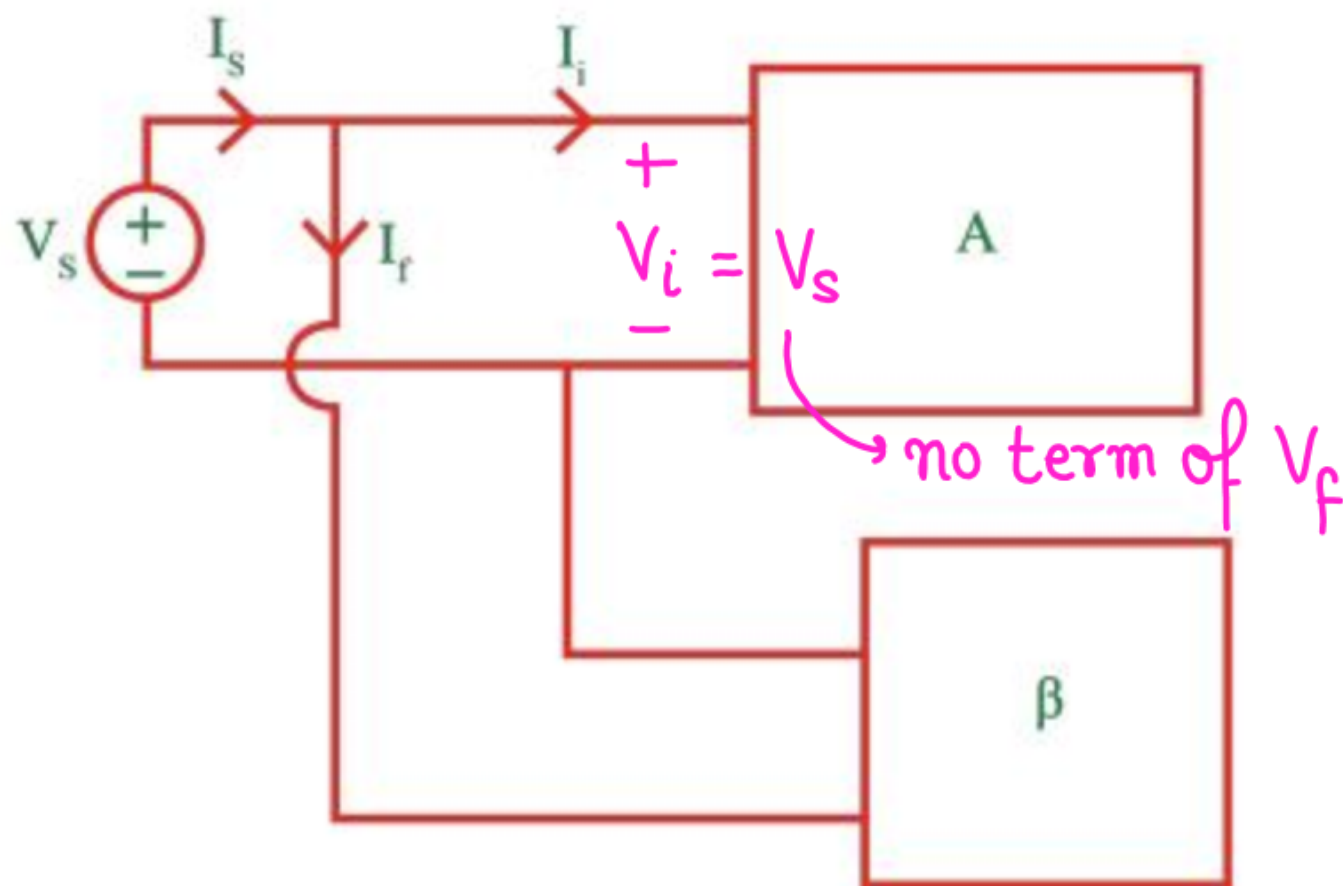


$$-V_f + V_s - V_i = 0$$

$$V_i = V_s - V_f : \text{voltage mixing}$$

(2) Shunt Mixing:

or current mixing



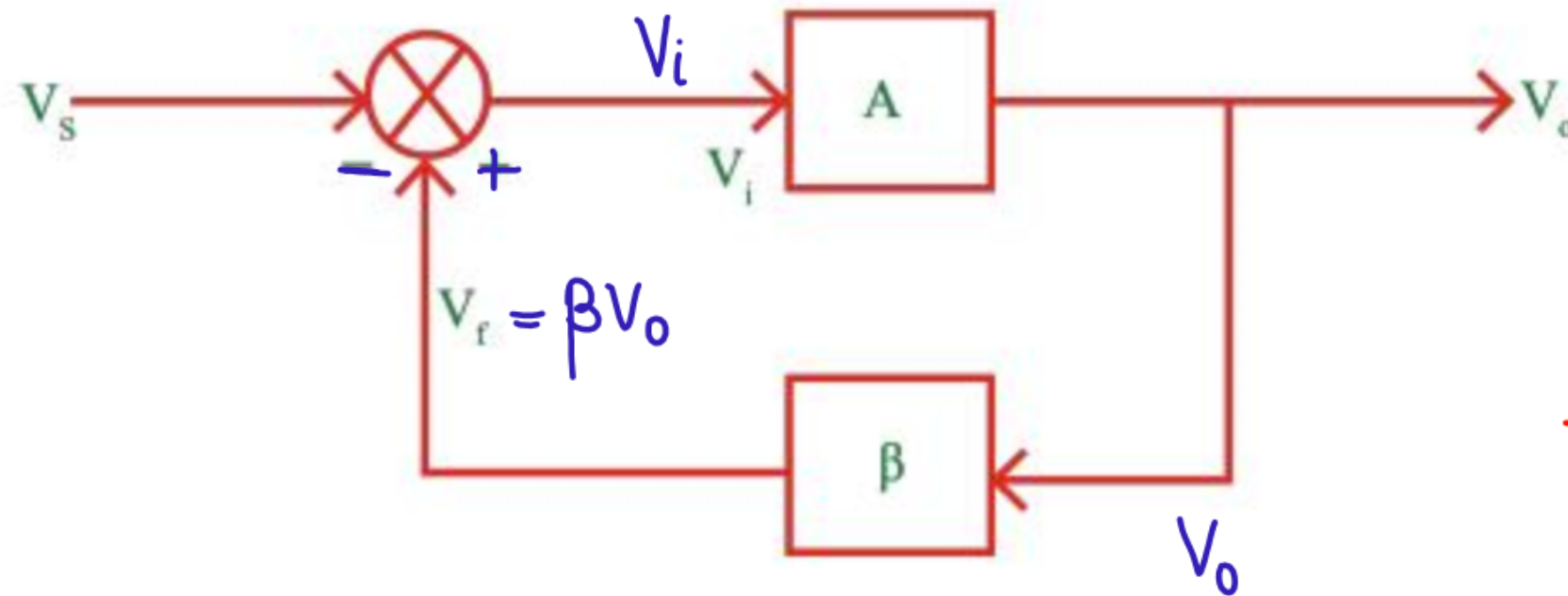
• The output of feedback n/w is connected parallel to source.

↳ Current input to amp^r is mix of source & feedback current.

$$\text{KCL: } I_i = I_s - I_f$$

↳ current mixing

Closed Loop Gain of an Amplifier :



$$V_i = V_s \mp \beta V_o$$

$$V_o = AV_i = AV_s \mp A\beta V_o$$

$$V_o (1 \pm A\beta) = AV_s$$

$$\text{closed loop gain } \frac{V_o}{V_s} = \frac{A}{1 \pm A\beta}$$

-ve feedback

$$A_{CL} = \frac{A}{1 + A\beta}$$

+ve feedback

$$A_{CL} = \frac{A}{1 - A\beta} \rightarrow \text{used in oscillators}$$

$$\text{loop gain} = A\beta$$

-ve feedback

$$A_f = \frac{A}{1 + A\beta} < A$$

feedback in dB

$$= 20 \log \left| \frac{A_f}{A} \right|$$

$$= 20 \log \left| \frac{1}{1 + A\beta} \right|$$

degenerative f/b

+ve feedback

$$A_f = \frac{A}{1 - A\beta} > A$$

feedback in dB

$$= 20 \log \left| \frac{A_f}{A} \right|$$

$$= 20 \log \left| \frac{1}{1 - A\beta} \right|$$

regenerative f/b

Gain Sensitivity :

1. Sensitivity indicated the change in closed loop gain of amplifier w.r.t the change in open loop gain .
2. Sensitivity of an amplifier should be as low as possible so that if open loop gain changes due to external environment ,the closed loop gain should remain same .

$$A_f = \frac{A}{1 + A\beta}$$
$$\frac{\partial A_f}{\partial A} = \frac{[1 + A\beta] - A(\beta)}{(1 + A\beta)^2}$$
$$= \frac{1}{(1 + A\beta)^2}$$

$$S = \frac{\text{fractional change in } A_f}{\text{fractional change in } A}$$
$$= \frac{dA_f/A_f}{dA/A} = \frac{A}{A_f} \frac{dA_f}{dA}$$
$$= (1 + A\beta) \times \frac{1}{(1 + A\beta)^2}$$
$$= \frac{1}{(1 + A\beta)}$$

no feedback

$$\beta = 0$$

$$S = \frac{1}{1 + A\beta} = 1$$

amount of varⁿ in
A = amount of varⁿ
in A_f

-ve feedback

$$\beta \neq 0$$

$$S = \frac{1}{1 + A\beta}$$

% change in A_f

$$= \frac{\% \text{ change in } A}{1 + A\beta}$$

varⁿ in A_f ↓

De-Sensitivity

$$D = \frac{1}{S} = (1 + A\beta)$$

S should be as low as possible or D should be as high as possible.

Effect of Negative Feedback on an Amplifier

1. Frequency Distortion

→ gain as a fn. of freq.

- If the amplifier contains reactive elements then gain is a function of frequency.
- Due to negative feedback the effect of change of open loop gain is minimized so frequency distortion is reduced.

◦ in any amp^r, gain varies with freq due to coupling, bypass & internal cap etc

◦ if β is realized using resistive n/w we can minimize varⁿ of gain with freq.

2. Non-Linear Distortion

- The non-linear distortion is reduced by a factor of $1+A\beta$

Noise

- Noise is signal which produce unwanted effect and degrade the quality of signal.

◦ due to negative feedback noise signal at o/p is reduced by a factor of $(1+A\beta)$.

$$V_o = AV_i + V_N + V_D$$

↘ distortion

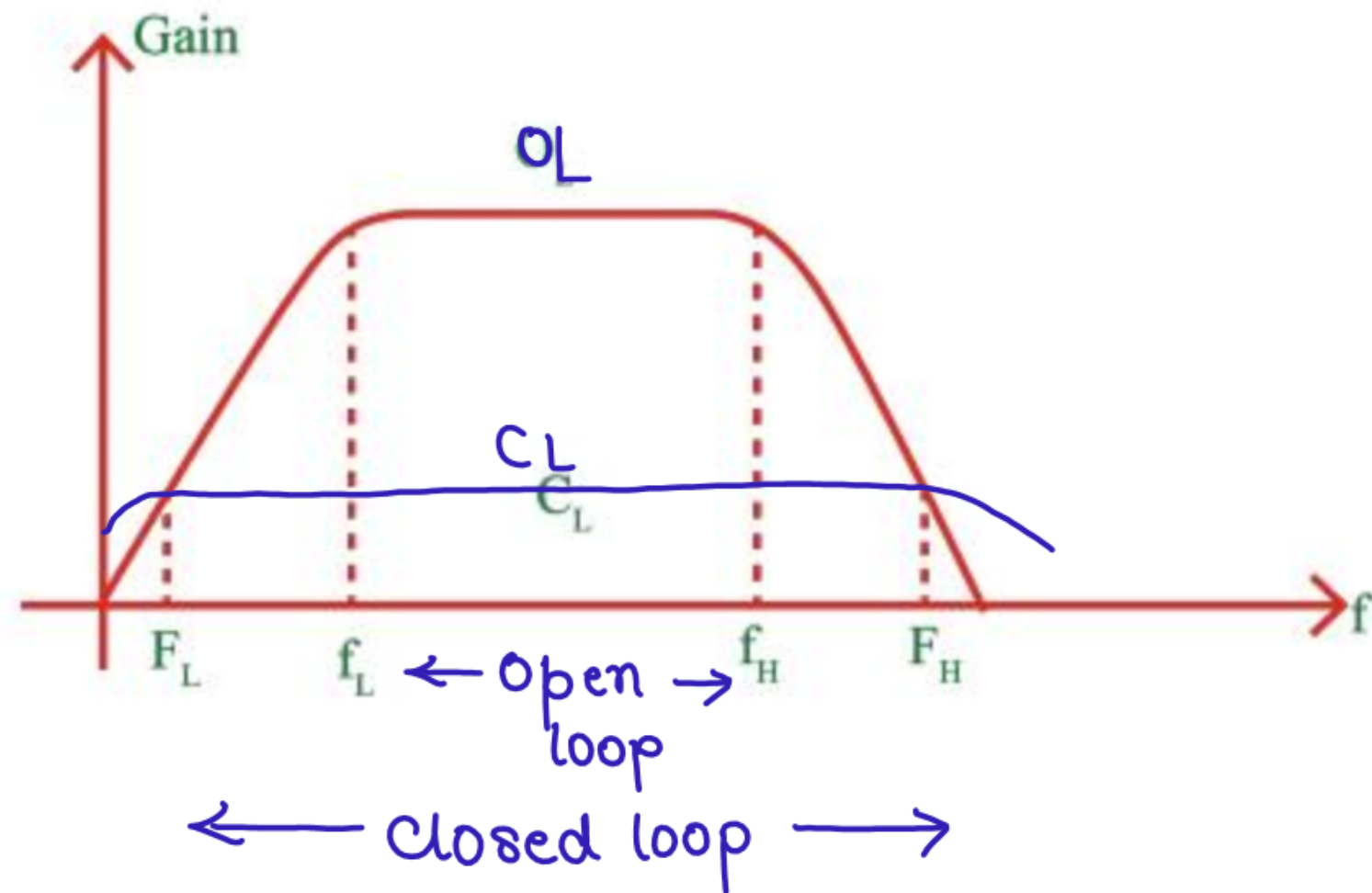
$$V_o = A(V_s - V_f) + V_N + V_D$$

$$V_o = AV_s - A\beta V_o + V_N + V_D$$

$$V_o(1+A\beta) = AV_s + V_N + V_D$$

$$V_o = \frac{AV_s}{1+A\beta} + \frac{V_N}{1+A\beta} + \frac{V_D}{1+A\beta}$$

BANDWIDTH



low freq gain

$$A_{LF} = \frac{A}{1 - j\left(\frac{f_L}{f}\right)}$$

$$\begin{aligned}(A_{LF})_f &= \frac{A_{LF}}{1 + A_{LF} \beta} \\ &= \frac{A / 1 - j\left(\frac{f_L}{f}\right)}{1 + \frac{A \beta}{1 - j\left(\frac{f_L}{f}\right)}}\end{aligned}$$

$$(A_{LF})_f = \frac{A}{1 - j\left(\frac{f_L}{f}\right) + A\beta}$$

$$= \frac{A/1+A\beta}{1 - j\left(\frac{f_L}{(1+A\beta)f}\right)}$$

$$= \frac{A_f}{1 - j\left(\frac{f_L}{f}\right)}$$

$$f_L = \frac{f_L}{1+A\beta} < f_L$$

High freq response

$$A_{HF} = \frac{A}{1 + j\left(\frac{f}{f_H}\right)}$$

$$(A_{HF})_f = \frac{A_{HF}}{1 + A_{HF}\beta}$$

$$= \frac{A/1+j\left(\frac{f}{f_H}\right)}{1 + \frac{A\beta}{1+j\left(\frac{f}{f_H}\right)}}$$

$$= \frac{A}{1 + j\left(\frac{f}{f_H}\right) + A\beta}$$

$$(A_{HF})_f = \frac{A/(1+A\beta)}{1 + j \left(\frac{f}{f_H(1+A\beta)} \right)}$$
$$= \frac{A_f}{1 + j \left(\frac{f}{f_H} \right)}$$

$$f_H = f_H(1+A\beta)$$

$$BW = f_H - f_L \approx f_H$$
$$= f_H(1+A\beta)$$

∴ gain reduces by a factor of $(1+A\beta)$
BW increases by a factor of $(1+A\beta)$

$$f_H = f_H(1+A\beta)$$

$$f_L = \frac{f_L}{1+A\beta}$$

• noise & distortion are reduced by a factor of $(1+A\beta)$.

FEEDBACK TOPOLOGIES

- ^{mixing} Series-Shunt or Voltage Series Feedback ^{samp.}
- ^{Samp.} Shunt-Series or Current Shunt Feedback ^{mixing}
- ^(I_o) Series-Series or Current Series Feedback ^(I_f)
- ^(V_f) Shunt-Shunt or Voltage Shunt Feedback ^(I_o)
- ^(I_f) Shunt-Shunt or Voltage Shunt Feedback ^(V_o)

mixing
Series - Shunt
(Voltage)

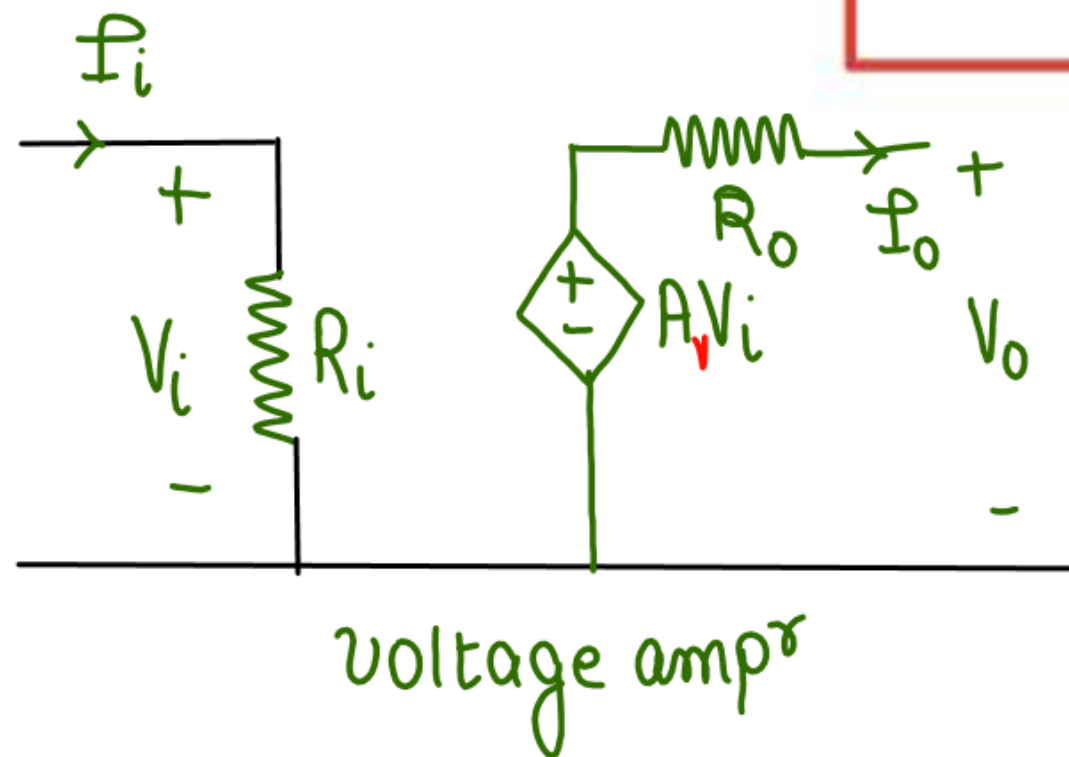
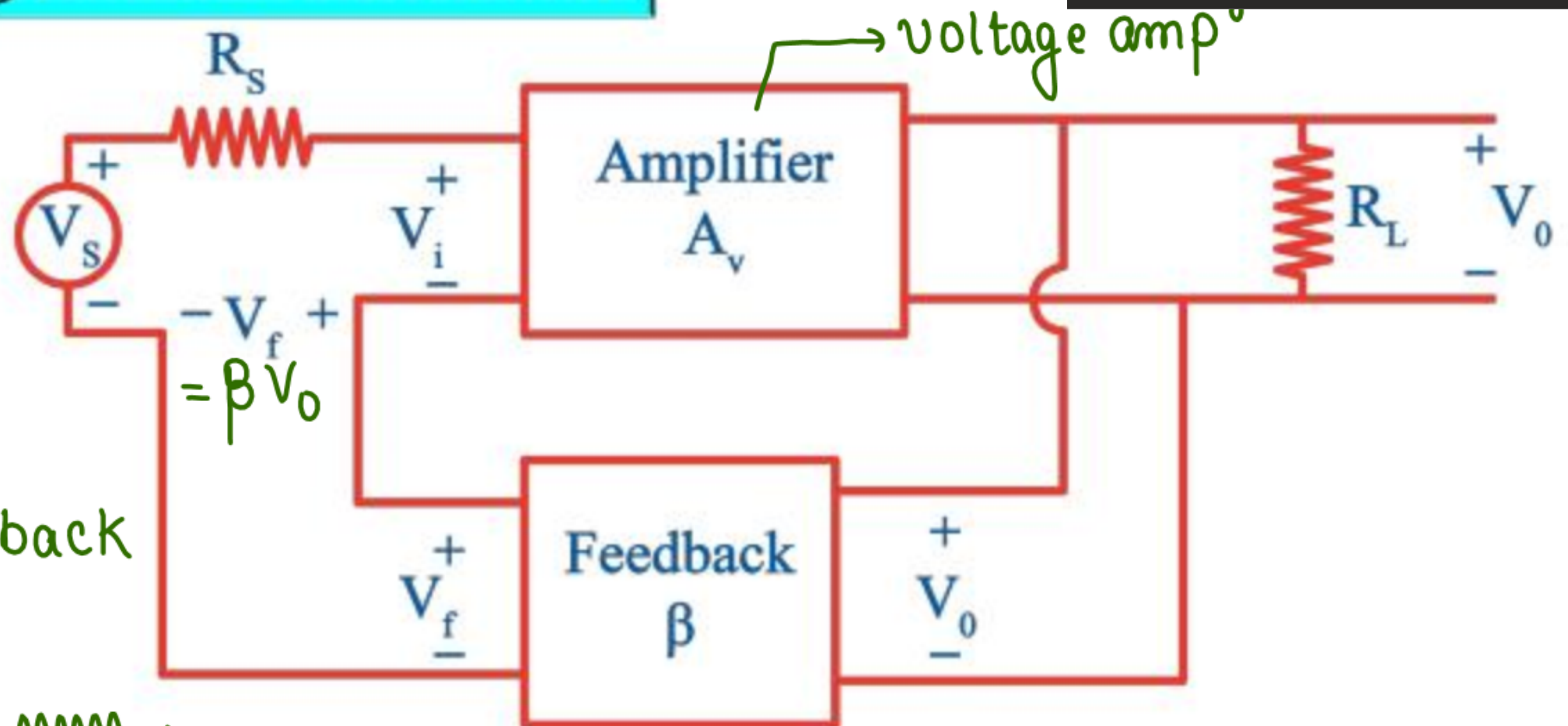
Voltage-Series Negative Feedback

↓ sampled V_o ↘ mixed V_f

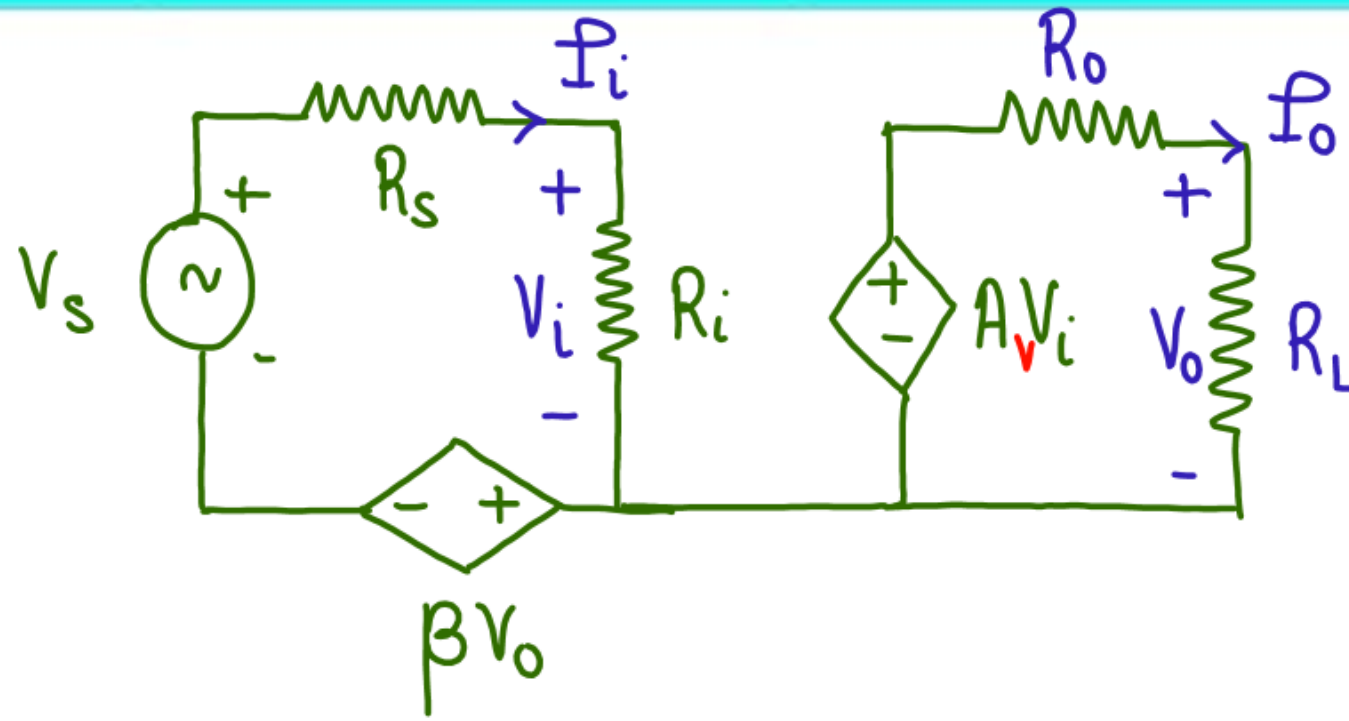
feedback factor

$$\beta = V_f / V_o \text{ (unitless)}$$

or Series-Shunt Feedback



Voltage-Series Negative Feedback



KVL in input loop

$$V_s = I_i R_s + I_i R_i + \beta V_o$$

$$V_o = A_v V_i \times \frac{R_L}{R_o + R_L}$$

$$\frac{A_v R_L}{R_o + R_L} = A_{vL}$$

$$V_s = I_i R_s + I_i R_i + \beta A_{vL} V_i$$

$$\begin{aligned} V_s &= I_i R_s + I_i R_i + \beta A_{vL} I_i R_i \\ &= I_i [R_s + R_i (1 + A_{vL} \beta)] \end{aligned}$$

input resistance with f/b

$$R_{if} = R_i (1 + A_{vL} \beta) > R_i$$

KVL in o/p loop

$$V_o = A_v V_i - I_o R_o$$

$$V_i = \frac{(V_s - \beta V_o) R_i}{R_s + R_i}$$

$$\begin{aligned} V_o &= \frac{A_v R_i}{R_s + R_i} (V_s - \beta V_o) - I_o R_o \\ &= A_{vs} V_s - A_{vs} \beta V_o - I_o R_o \end{aligned}$$

$$A_{vs} = \frac{A_v R_i}{R_s + R_i}$$

$$V_o [1 + A_{vs} \beta] = A_{vs} V_s - I_o R_o$$

$$V_o = \frac{A_{vs}}{1 + A_{vs} \beta} V_s - \frac{I_o R_o}{1 + A_{vs} \beta}$$

R_{of} = o/p resistance with f/b

$$= \frac{R_o}{1 + A_{vs} \beta} < R_o$$

$$I_o = V_o / R_L$$

$$V_o \left[1 + \frac{R_o}{R_L (1 + A_{vs} \beta)} \right] = \frac{A_{vs}}{1 + A_{vs} \beta} V_s$$

$$\begin{aligned} \text{gain} &= V_o / V_s \\ &= \frac{A_{vs} R_L}{R_o + R_L (1 + A_{vs} \beta)} \end{aligned}$$



Question-01

In a negative feedback amplifier using voltage series (i.e. voltage-sampling series mixing) feedback. (R_i and R_o denote the input and output resistance respectively)

- (a) R_i decreases and R_o decreases
- (b) R_i decreases and R_o increases
- ☒ (c) R_i increases and R_o decreases
- (d) R_i increases and R_o increases

voltage - series
shunt sampling series mixing
shunt at o/p series at i/p
 $R_o \downarrow$ $R_i \uparrow$



Question-02

A current controlled current source (CCCS) has an input impedance of $10\ \Omega$ and output impedance of $100\text{k}\Omega$. When this CCCS is used in a negative feedback closed loop with a loop gain of 9, the closed loop output impedance is

- (a) $100\text{ k}\Omega$
- (b) $100\ \Omega$
- (c) $10\ \Omega$
- ☒ (d) $1000\text{k}\Omega$

$A\beta$
Current controlled Current Source
 I_i I_o
shunt mixing series sampling
 $R_o \uparrow$

$$R_{of} = R_o(1 + A\beta) \\ = 100(1 + 9)$$



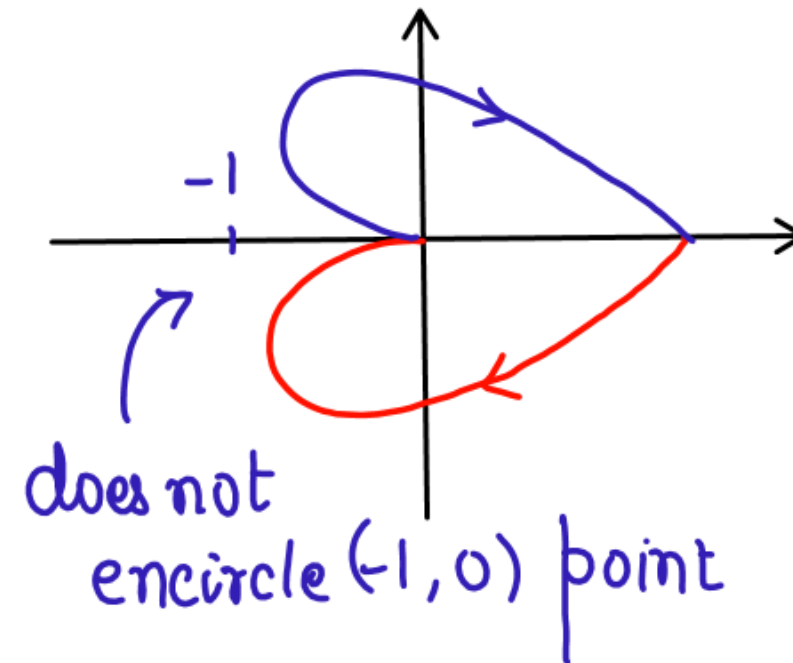
Question-03

→ β or H does not vary with f

An amplifier with resistive negative feedback has two left half-plane poles in its open-loop transfer function. The amplifier.

- (a) Will always be unstable at high frequencies
- ✓ (b) Will be stable for all frequencies
- (c) May be unstable, depending on the feedback factor
- (d) Will oscillate at low frequencies

$N = 0$
OL poles in LHP, $P = 0$
 $Z = P - N = 0$
no CL poles in RHP





Question-04

Negative feedback in an amplifier

- (a) Reduces gain
- (b) Increases frequency and phase distortions (reduces)
- (c) Reduces bandwidth (inc.)
- (d) Increase noise (reduce)



Question-05

Series-series feedback results in

- (a) Input impedance increases and output impedance decreases
- ☒ (b) Both input and output impedance increases
- (c) Input impedance decreases and output impedance increases
- (d) Both input and output impedance decreases



Question-06

If an amplifier with gain of -1000 and feedback of $\beta = -0.1$ has a gain change of 20 % due to temperature, then calculate the change in gain of the feedback amplifier -

- ☒ (a) 0.2 %
- (b) 0.02 %
- (c) 0.002 %
- (d) 2 %

$$\begin{aligned}\% \text{ change in } A_f &= \frac{\% \text{ change in } A}{1 + A\beta} \\ &= \frac{20\%}{1 + (-1000)(-0.1)} \approx \frac{20}{100}\% \\ &= 0.2\%\end{aligned}$$



Question-07

In an amplifier with a gain of 1000 without feedback and cut-off frequencies at 2 kHz and 20 kHz, negative feedback of 1 % is employed. The cut-off frequencies with feedback would be

$$\beta = 0.01$$

- (a) 220 Hz and 22 kHz
- ☒ (b) 182 Hz and 220 kHz
- (c) 220 Hz and 220 kHz
- (d) 182 Hz and 22 kHz

$$f_L = \frac{f_L}{1 + A\beta} = \frac{2\text{ K}}{1 + 1000 \times 0.01} = \frac{2000}{11} = 181.8\text{ Hz}$$

$$f_H = f_H (1 + A\beta) = 20\text{ K} \times 11 = 220\text{ KHz}$$



Question-08

An amplifier circuit has an overall current gain of -100 and an input resistance of $10\text{ k}\Omega$ with a load resistance of $1\text{ k}\Omega$. The overall voltage gain of the amplifier is

- (a) 5 dB
- (b) 10 dB
- ☒ (c) 20 dB
- (d) 40 dB

$$\begin{aligned} A_v &= \frac{A_i R_L}{R_i} = \frac{I_o R_L}{I_i R_i} \\ &= \frac{100 \times 1\text{K}}{10\text{K}} = 10 \\ 20 \log 10 &= 20\text{ dB} \end{aligned}$$



Answers

Question No.	Answer
1	.
2	.
3	.
4	.
5	.
6	A
7	B
8	C



Student's Assignment

Question-01

In a negative feedback amplifier using voltage series (i.e. voltage-sampling series mixing) feedback. (R_i and R_o denote the input and output resistance respectively)

- (a) R_i decreases and R_o decreases
- (b) R_i decreases and R_o increases
- (c) R_i increases and R_o decreases
- (d) R_i increases and R_o increases



Question-02

In a shunt-shunt negative feedback amplifier, as compared to the basic amplifier

- (a) Both input and output impedances decrease
- (b) Input impedance decreases but output impedances increases
- (c) Input impedance increases but output impedances decreases
- (d) Both, input and output impedances increase



Question-03

Voltage series feedback (also called series shunt feedback) result in

- (a) Increase in both input and output impedances
- (b) Decrease in both input and output impedances
- (c) Increase in input impedance and decrease in output impedances
- (d) Decrease in input impedance and increase in output impedances



Question-04

Statement (I) : In a negative feedback amplifier, the noise voltage due to noise generated within the amplifiers is reduced

Statement (II) : Negative feedback reduced the amplifier gain

- a. Both statement (I) and Statement (II) are individually true and Statement (II) is the correct explanation of Statement (I)
- b. Both Statement (I) and Statement (II) are individually true but statement (II) is not the correct explanation of Statement (I)
- c. Statement (I) is true but Statement (II) is false
- d. Statement (I) is false but Statement (II) is true



Question-05

The effect of negative feedback is to increase the of a series voltage negative feedback amplifier by a factor of $(1+A_v \beta)$.

- (a) voltage gain
- (b) output impedance
- (c) input impedance
- (d) bandwidth



Question-06

A feedback amplifier has an open loop gain of -100 . If 4% of the output is fed back in a degenerative loop, what is the closed loop gain of the amplifier?

- (a) -33.3
- (b) -25
- (c) -20
- (d) $+25$



Question-07

Applications of negative feedback to a certain amplifier reduced its gain from 200 to 100 . If the gain with the same feedback is to be raised to 150 , in the case of another such appliance, the gain of the amplifier without feedback must have been

- (a) 400
- (b) 450
- (c) 500
- (d) 600



Question-08

An amplifier has a voltage gain of 120 . To reduce distortion, 10 % negative feedback is employed. The gain of the amplifier with feedback is

- (a) 141
- (b) 92.3
- (c) 9.23
- (d) 1.41



Answers

Question No.	Answer
1	
2	
3	
4	A
5	D
6	.
7	D
8	C



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